Aerial Robotics Lab

Imperial College London

3rd Year Group Design Project
Bio-inspired Aerial Construction

Extrusion System Design

BAC 09b

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Abstract

Faculty of Engineering
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Masters in Aeronautical Engineering

Extrusion System Design

by Robert WATKISS

In order to facilitate autonomous aerial construction an extrusion system was designed to deposit material and build a pre-defined structure measuring 50cm high. A systems engineering approach was used to decompose the problem into its constituent parts using functional analysis in a top-down manner. Individual subsystems were designed and developed from a variety of potential solutions before a bottom-up integration procedure. The resulting extrusion system was capable of translating G-code strings into motion, extrusion and perching commands in order to control the full extrusion process. To the best of the author's knowledge, the system presented in this report represents a novel method of 3D printing that has never been successfully achieved before.

Acknowledgements

I would like to thank my team-mate and friend, Mr Marcus Rose, for his unending enthusiasm throughout this project and his incessant drive for perfection. My gratitude also extends to the workshop staff, in particular Mr Roland Hutchins, for their help and guidance. Their knowledge and experience proved invaluable.

Thanks also go to the project supervisors, particularly Dr Nigel MacCarthy, for their feedback and support allowing us to learn through the course of this project. Finally I would like to thank Dr Mirko Kovac for his leadership and knowledge in the area of flying robotics. Without him this project would have not been possible.

1 Introduction

1.1 Motivation

There are many things that nature can do with apparent ease and finesse which we are as yet unable to recreate with man-made technology. A swallow, for example, can fly with agility and grace, perch on seemingly smooth surfaces and build it's own nest from nothing but mud [1]. To replicate these abilities would allow for autonomous construction in environments too hazardous for humans or for the building of more complex structures than would otherwise be possible. The purpose of this group design project is to design, build and test a bio-inspired aerial construction vehicle (ACV).

FIGURE 1.1: A swallows nest built from mud against a concrete structure [1]

1.2 Literature Review

Autonomous robotic assembly and construction has seen increasing prevalence since the early 1980s [2] and is typically used to replace human workers where the task in question is repetitive, dull or requires high precision [3]. Robots have been implemented in systems requiring multiple agents in order to achieve tasks that would otherwise be impossible, such as a pair of autonomous bridge building robots [4].

Whilst robotic construction on terra-firma is commonplace, the field of aerial construction is still in its infancy. To date, the only truly aerial construction vehicles have been quadrotors extruding two-part polyurethane foam [5] and swarms of quadrotors assembling blocks and struts into a simple cubic structure [6].

Looking to nature, spiders are capable of self-secreting a building material in order to construct intricate webs [7] and birds are capable of assembling nests from twigs [8] or just mud [1].

Drawing insight from such creatures using the inspire-abstract-implement bio-inspiration paradigm [9] could pave the way for future autonomous aerial construction.

1.3 Problem Definition

In order to facilitate autonomous printing, the ACV needs to deposit sufficient material to build a large structure in a short period of time. An extrusion mechanism mounted to the ACV must be able to continuously and reliably print for the duration of the mission without needing to be reset or restocked.

1.4 Objectives

The objective of the extrusion system design was to design, build and test a system that can satisfy the problem definition. The solution must

- process the raw material into an extrudable substance,
- deposit material in a controlled and predictable manner through an *end effector*,
- allow the *end effector* to follow a toolpath and produce complex structures,
- compensate for motion of the ACV and keep the *end effector* on the defined toolpath,
- control the perching mechanism to allow for z-axis authority during extrusion.

2 System Architecture

By employing systems engineering methods [10], the extrusion system was decomposed into it's constituent parts. This allowed us to tackle the problem in a systematic, coordinated manner whilst considering the interrelations between different subsystems.

2.1 System Decomposition

The extrusion system was broken down into two primary subsystems; the extruder system and the motion system. The extruder system must store, pump, mix and extrude the printing material whilst the motion system must move and control the end effector. This is summarised in Figure 2.1, below. Within our team, Mr Marcus Rose initially focussed on the extruder system whilst my role was to develop the motion system and develop the extrusion system control scheme as a whole.

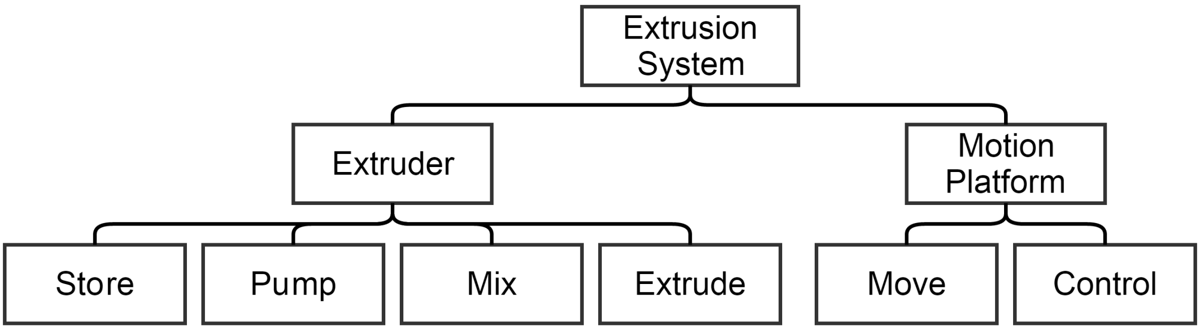


FIGURE 2.1: Extrusion system diagram decomposed by function

2.2 Proposed System Architecture

Figure 2.2, below, shows the proposed extrusion system architecture. This is a hierarchical diagram showing the interconnection of different systems and subsystems on the Aerial Construction Vehicle. The flight control subsystems (1.1.x) have been excluded for clarity. The perching system (1.2.1.1) was designed by another group, BAC10, however was implemented under the extrusion control system so has been included for completeness.

We decided to use two separate onboard processors for the flight control (1.1) and extrusion control (1.2) systems as this allowed for independent, and hence less complex, development. A CRIUS AIOP v1 [11] development board was chosen as it allowed for easy component connections (using three-pin servo wires) and it uses the well documented ARDUINO framework [12]. Furthermore, this controller has onboard 9-Axis IMU sensors which would allow for basic end effector stabilisation.

A G-code interpreter translates standard G-code strings [13] into perching, motion and extruder commands. The syntax and interpretation of G-code is beyond the scope of this report however it is well documented online and in [13]. Both the perching system (1.2.1.1) and extruder system (1.2.1.3) required DC motor drive with current draws exceeding the limits of the control board. We designed a control circuit using MOSFETs, a type of transistor [14], to drive these motors. In the case of the perching motor (1.2.1.1.1.1) it was required to drive the motor both forwards an backwards. A MOSFET H-Bridge was implemented to achieve this [15].

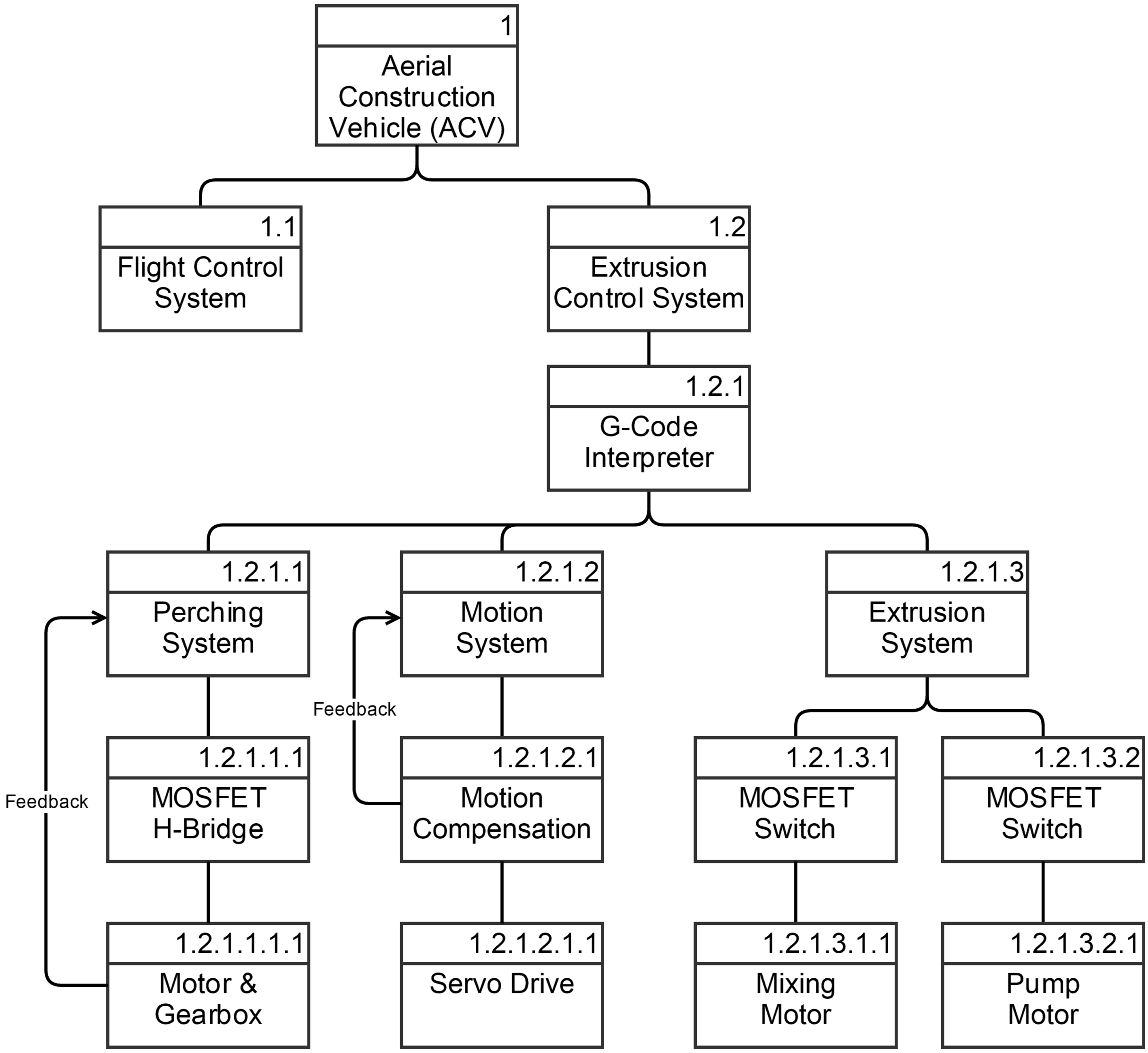


FIGURE 2.2: Proposed extrusion system architecture

3 Motion System Design

The motion system was responsible for moving the end effector in order to follow a toolpath and compensate for the motion of the ACV during printing.

3.1 Design Requirements

In order to satisfy the problem definition, the motion system solution must

- allow the end effector to follow a toolpath and produce complex structures,
- compensate for spacial inaccuracies of the ACV during construction,
- be lightweight to allow for practical use on an ACV,
- have a spatial resolution greater than that of the material.

3.2 Proposed Solutions

There are a number of existing solutions for motion systems, each with their own benefits and limitations. We considered the use of a Cartesian Platform (Fig 3.1(a)), a Polar Gimbal (Fig 3.1(b)), a traditional Robotic Arm (Fig 3.1(c)) and a Delta Robot (Fig 3.1(d)).

Cartesian Platform

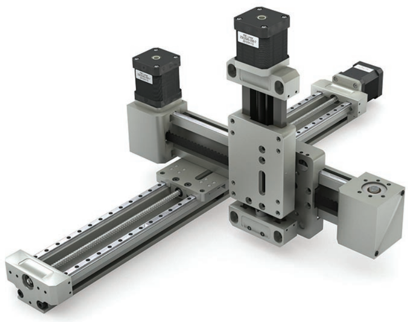
A Cartesian platform, commonly seen on CNC equipment such as plotters, routers and conventional 3D printers uses two or three independently controlled axes for authority in 2- or 3- dimensional space [20]. These axes are typically controlled by stepper motors driving lead screws or belts. They are commonplace since they can achieve very high resolution [20] however this is not a critical design factor for our motion system due to the scale of construction we are considering. The design is slow, heavy and has a significant moving mass which is undesirable for an ACV.

Polar Gimbal

A polar gimbal controls yaw and pitch using two servomotors, effectively giving it authority over R-θ space. By extending an end effector from this gimbal and coupling control with the perching system it is possible to reach any point in 3D space with very little moving mass. Relying on the perching system for vertical motion was undesirable since it was expected to be slow and require significant power draw.

Robot Arm

With a robot arm it is possible to place the end effector at any orientation in 3D space, effectively achieving 6-dimensional control authority. This is extremely impressive however unnecessary at this stage of ACV development. Due to the *parent-child* design of a robot arm there is a significant moving mass and the design, in general, is reasonably heavy and slow.



(a) Cartesian Platform [16]

(b) Polar Gimbal [17]

(c) Robotic Arm [18]

(d) Delta Robot [19]

FIGURE 3.1: Proposed motion system solutions

Delta Robot

A Delta robot is a type of parallel robot which uses three motors to place an end effector anywhere in 3D space [21]. The use of parallelograms in the arm design means that the end effector is always parallel to the robot base. A key design feature of a delta robot is that the heavy driving motors are all stationary and mounted at the centre of mass meaning there is minimal change of inertia during operation. This was particularly beneficial since it makes the delta robot very fast and eases stabilisation of the ACV.

3.3 Comparison of Designs

By defining and weighting the desirable features of the motion system we were able to select the most suitable solution for our design. As shown in Table 3.1, the Delta Robot was identified as the most likely to satisfy the problem definition and was taken forward to design and development.

Feature	**Weighting (1-5)**	**Cartesian Platform**	**Polar Gimbal**	**Robot Arm**	**Delta Robot**
Small Moving Mass	5	✗	✓	✗	✓
Lightweight	5	✗	✓	✗	✓
High Resolution	2	✓	○	✓	○
3D-Space Authority	3	✓	✗	✓	✓
Speed	4	○	✗	○	✓
Simplicity of Design	4	✓	✓	✗	✓
		111	143	79	**199**

TABLE 3.1: Comparison matrix for different motion system solutions
(✓=9, ○=5, ✗=1)

3.4 Motion System Implementation

A simple delta robot was designed based on freely available open source designs [22] using low cost hobby parts. The result is shown in Figure 3.2. The base of the robot, 1, was laser cut from 3mm plywood and sandwiches three TURNIGY TGY-9018MG [23] metal gear servomotors, 2. This is held together using simple cable ties. Clamped to each of these servos is an arm, 3, which consists of a *shoulder-to-elbow* strut, a very basic ball joint and a pair of *elbow-to-wrist* struts, 4. The three arms connect to the end effector, 5, using another set of ball joints. Excluding fasteners, the parts of subassemblies 3 and 5 were 3D printed. M3 sized threaded rod was used for 4. The designs for these parts accompanies this report.

The delta robot was controlled using a modified version of an open source inverse kinematics algorithm [22]. A full explanation of inverse kinematics is beyond the scope of this report however it is described in [24]. The ARDUINO code accompanies this report.

To verify the robot design against the design requirements we ran a series of simple tests. We confirmed that the robot was able to produce both large and small movements very quickly and with a reasonable resolution. A video of this test accompanies this report. Three screenshots are shown in Figure 3.3.

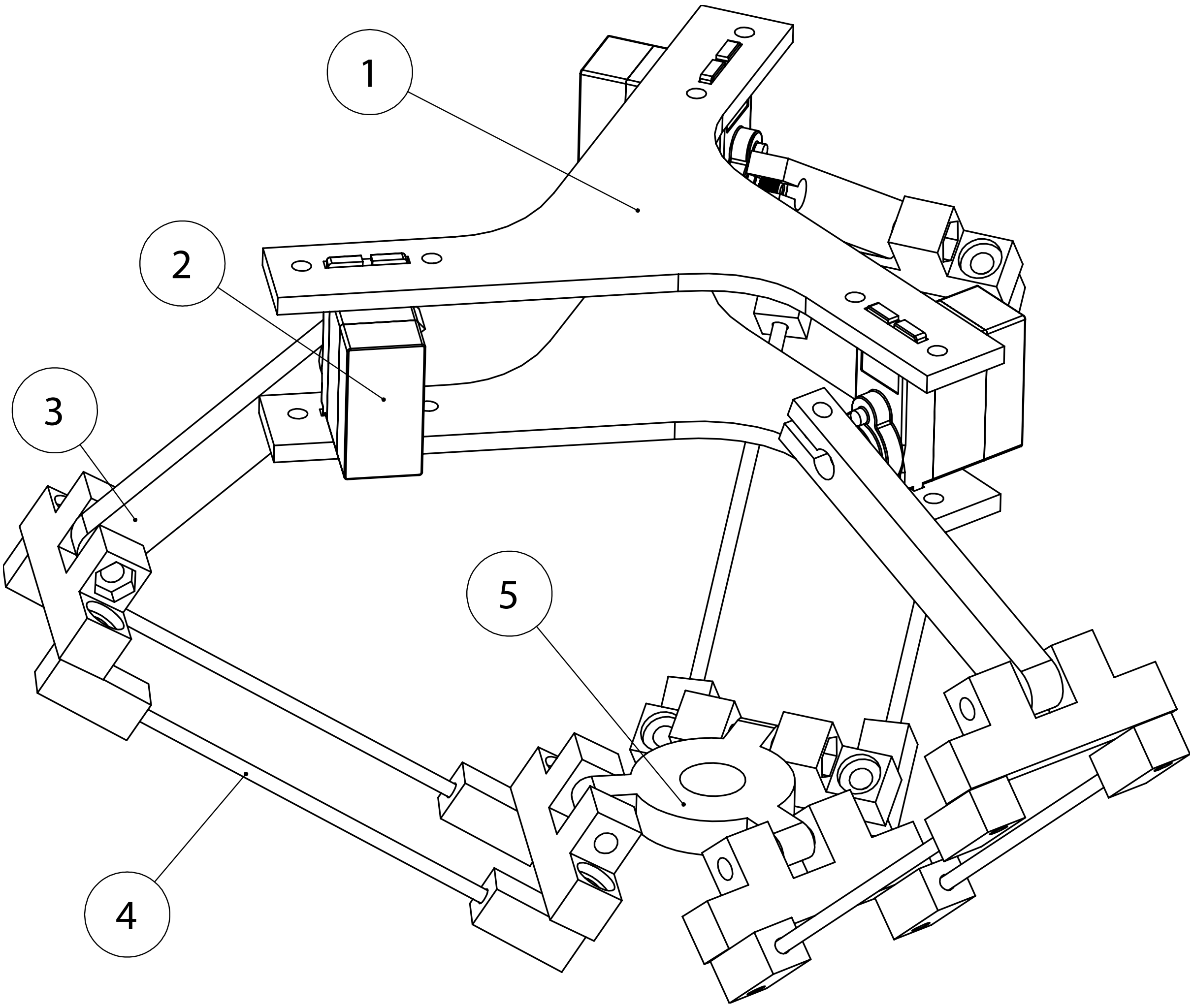


Figure 3.2: Delta robot design

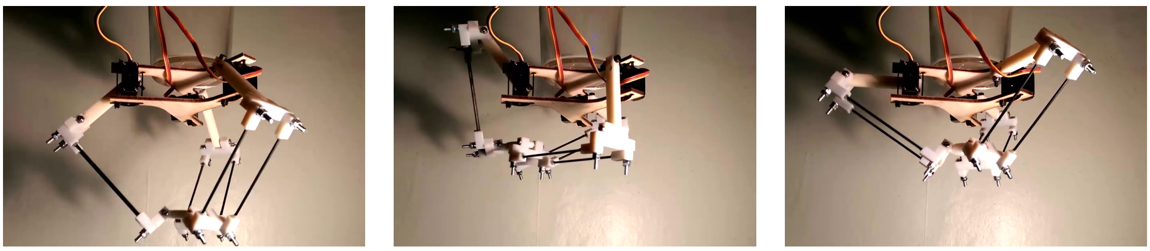


Figure 3.3: Screenshots of delta robot performance verification testing

4 Extruder System Design

The extruder system was responsible for processing and depositing raw material in order to build a structure. The system was made up of four constituent subsystems; storage, pumping, mixing and extrusion.

4.1 Design Requirements

In order to satisfy the problem definition, the extruder system must

- store sufficient raw material to build a structure approximately 50cm in height,
- process the raw material into an extrudable substance,
- deposit material in a controlled and predictable manner,
- be capable of operating continuously for the entire construction process,
- be lightweight in order to minimise the moving mass and allow for mounting on an ACV.

4.2 Material Characteristics

After extensive testing of various different materials, groups BAC05 and BAC06 chose a two-part polyurethane foam as the build material. This is a closed cell foam commonly used for insulating buildings or constructing watercraft [25]. The raw material needs to be mixed for approximately 90 seconds to achieve the optimum properties and expansion ratio. Full expansion is achieved at approximately 3 minutes and it is hardened by 6 minutes. It is worth noting that the raw materials are reasonably viscous as this influenced our design choices.

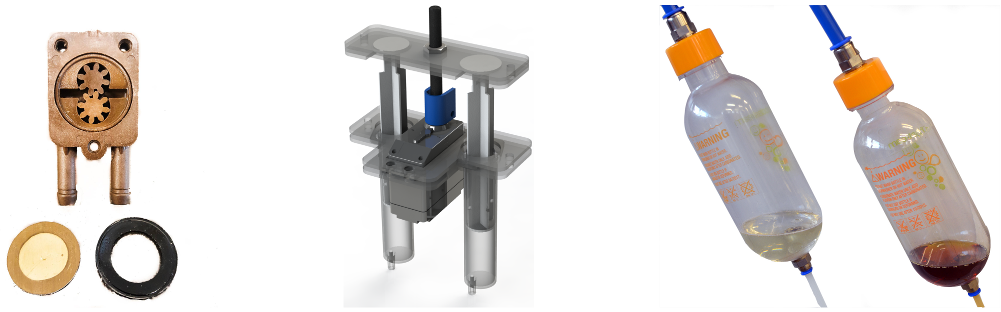
4.3 Pumping Subsystem

In order to allow the extruder system to operate continuously the design required the two components of the expanding foam to be kept separate as long as possible. This stipulated the requirement that the two parts must be pumped individually whilst maintaining a 1:1 mixture and hence equal flow rates.

Three different pumping mechanisms were identified and trialled, these were gear pumping, direct piston pumping and pneumatic pumping.

Gear Pumping

Gear pumps, such as that shown in Figure 4.1(a), are often used for pumping viscous liquids such as oil or coolant [26]. They use the meshing of two gears to drive fluid. For our initial testing we disassembled a fuel pump intended for a model plane engine. We were dissatisfied when attempting to pump water however it was significantly more effective



(a) Gear Pump (b) Piston Pump (c) Pressurised Bottles

FIGURE 4.1: Proposed pumping mechanisms.

at pumping the more viscous raw material. Testing showed that the two expanding foam components had an equal flow rate, as desired.

Piston Pumping

Similar to the design previously used in an ACV mounted extruder [5], we trialled a piston pumping method, shown in Figure 4.1(b). This consisted of two syringes driven by a lead screw mounted on a continuous rotation servomotor. Downstream of the piston pump we mounted a pair of two-way valves connecting the pistons, storage tanks and mixing subsystem which allowed for continuous operation without manual refilling. Testing was initially promising as we were able to achieve high pressure and equal flow rates however the system is inherently complex and proved to be unreliable.

Pneumatic Pumping

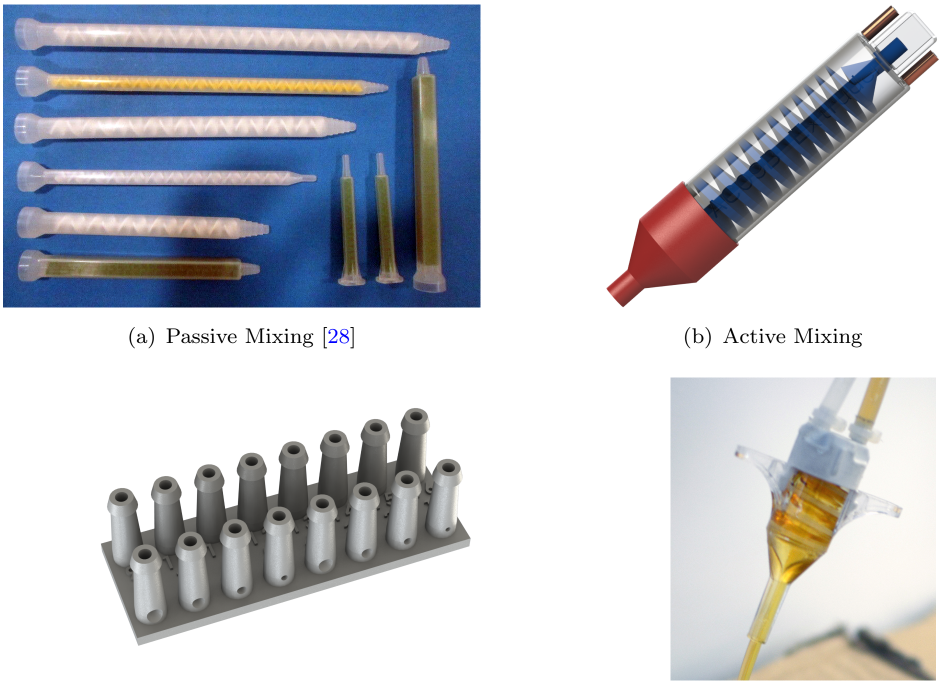
Inspired by the high pressure commercial solutions such as [27] we trialled a pneumatic pumping system. Two pressure vessels, shown in Figure 4.1(c), were partially filled with the two separate expanding foam components and pressurised to 10 bar. This proved to be very effective however the flow rate was hard to control and far in excess of our needs.

4.4 Mixing Subsystem

Testing by groups BAC05 and BAC06 showed that hand mixing for 90 seconds was required to produce the optimum material properties and expansion. In order to replicate and automate this we identified and trialled four different solutions. These were passive mixing, active mixing, aerodynamic mixing and high pressure nozzle mixing.

Passive Mixing

As an initial test we trialled driving the two part foam through commercially available epoxy mixing nozzles, such as those in Figure 4.2(a) [29]. This nozzle consisted of a series of bidirectional elements that force the two parts into contact in what is known as cyclonic mixing. The raw material, however, is significantly less viscous than two part epoxy and as a result they failed to mix sufficiently. By orientating the nozzle close to the horizontal



(a) Passive Mixing [28]

(b) Active Mixing

(c) Aerodynamic Mixing

(d) High Pressure Nozzle Mixing

FIGURE 4.2: Proposed mixing mechanisms

we were able to slow the flow through the nozzle and hence increase mixing time. The resulting mixture dripped out as a liquid and quickly formed a foam. We were able to produce a structure using this method to repeatedly drip on a rectangular path over a 50 minute test. An image of this structure is included in the appendices.

Active Mixing

In an attempt to accelerate the mixing process we designed an active mixing nozzle, shown in Figure 4.2(b). This used an auger to drive the two part foam whilst maximising mixing time. By empirically matching the material flow rate and auger speed to the expansion rate of the foam we were able to extrude a foam suitable for printing after approximately 90 seconds. Our initial designs, however, suffered from clogging issues and were limited to about three minutes of extrusion at a time.

Aerodynamic Mixing

Inspired by fuel injectors and pneumatic spray guns we trialled an aerodynamic mixing method using the rig shown in Figure 4.2(c) [30, 31]. The aim of this was to produce a spray of each component and direct them tangentially at each other. By maximising the surface area to volume ratio we expected to increase the rate of reaction and hence decrease the mixing time. Our tests were ultimately unsuccessful as we could not produce a spray of either component. We believe this is due to the high viscosity of the raw material. [32] suggests that heating the fluid or driving with a higher pressure would facilitate atomisation and potentially allow this method. Unfortunately neither of these are viable for an ACV mounted extruder at this stage.

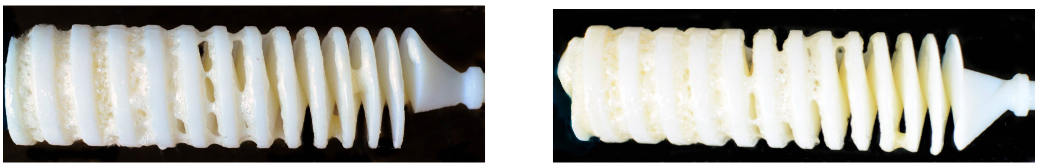
High Pressure Nozzle Mixing

In conjunction with the pneumatic pumping system described above, we trialled a commercial mixing nozzle intended for spraying expanding foam for insulation [27]. This is shown in Figure 4.2(d). During testing, 20ml of each component was driven through the nozzle at 10 bar over the course of seven seconds. This produced a solid structure approximately 8000cm^3 in volume after 30 seconds of expansion. Whilst thoroughly mixed, the first material extruded had not started foaming and as such was not suitable for printing. Towards the end of the test, however, the fluid level in the tank dropped to the point that air was also drawn through the nozzle and the material extruded had already started foaming. We believe this mimics the blowing agent found in commercial solutions and is a potential path of future research.

4.5 Extrusion System Implementation

Based on the trialled solutions for the pumping and mixing subsystems we decided that the active mixing method with gear pumps was the most viable system based on our design requirements. The storage subsystem simply consisted of unpressurised tanks designed to fit around the ACV frame and the extruder subsystem was a tapered nozzle fitted to the output of the mixing subsystem.

To overcome the clogging issues faced in initial testing of an active mixer we identified five key design variables that influenced the extruded material properties. These were the auger length, pitch and speed, the raw material flow rate and the nozzle diameter. We found that the optimum characteristics were achieved when the auger length, pitch and speed resulted in a total of 90 seconds mixing. By varying the pitch of the auger along it's length to match the material expansion we were able to prevent any back flow and hence clogging. Comparing the auger after five minutes of operation (Figure 4.3(a)) and thirty minutes (Figure 4.3(b)) there is very little build up of material over time suggesting that continuous operation beyond this is unlikely to be problematic. A more detailed explanation of the active mixing mechanism can be found in the report by Mr Marcus Rose.



(a) Auger after two minutes

(b) Auger after 30 minutes

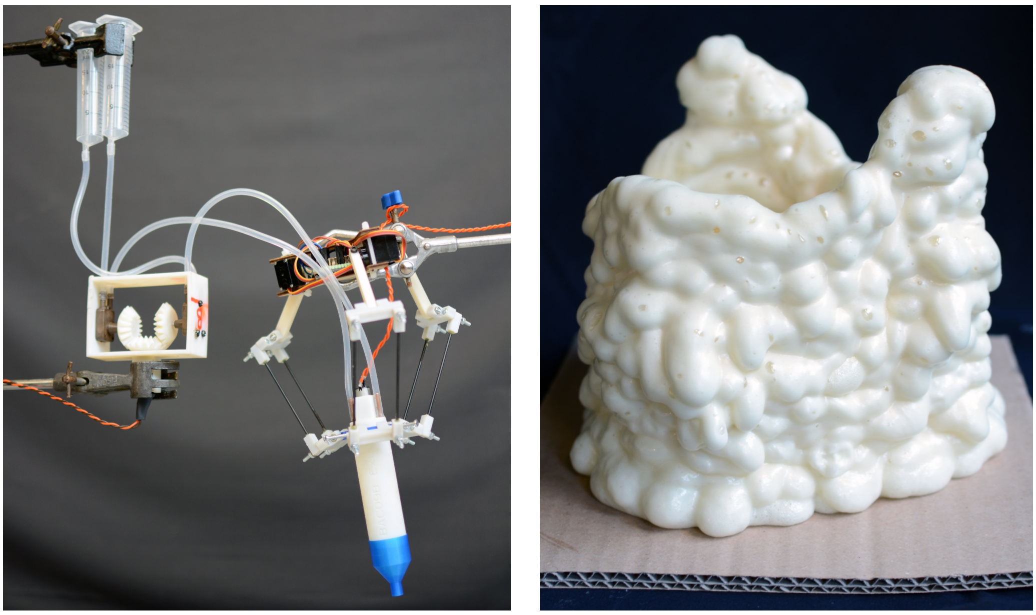
FIGURE 4.3: Comparison of material deposition on auger after two and 30 minutes

5 System Implementation

5.1 Bench Testing

To verify and validate the motion and extruder systems we conducted a series of bench tests. This allowed us to prove the system as a whole and refine it's performance. The setup, shown in Figure 5.1(a), consisted of two open syringes serving as tanks feeding the raw material to two geared pumps which in turn drive fluid into the active mixer mounted on a delta robot. To the best of our knowledge this method of 3D printing is novel and has never been successfully achieved before.

For the first print the motion system was instructed to trace a circle of diameter 150mm whilst the extruder system ran at a constant rate. The resulting structure, shown in Figure 5.1(b), is the result of a 29 minute print and was extremely promising. A timelapse video of this test accompanies this report. The three peaks occurred where the delta robot's servomotors reached their physical limits and are a result of attempting to print such a large structure. In further tests a structure with diameter 120mm was identified to be the limit of this system.



(a) Experimental setup

(b) Resulting structure

Figure 5.1: Bench test experimental setup and resulting structure

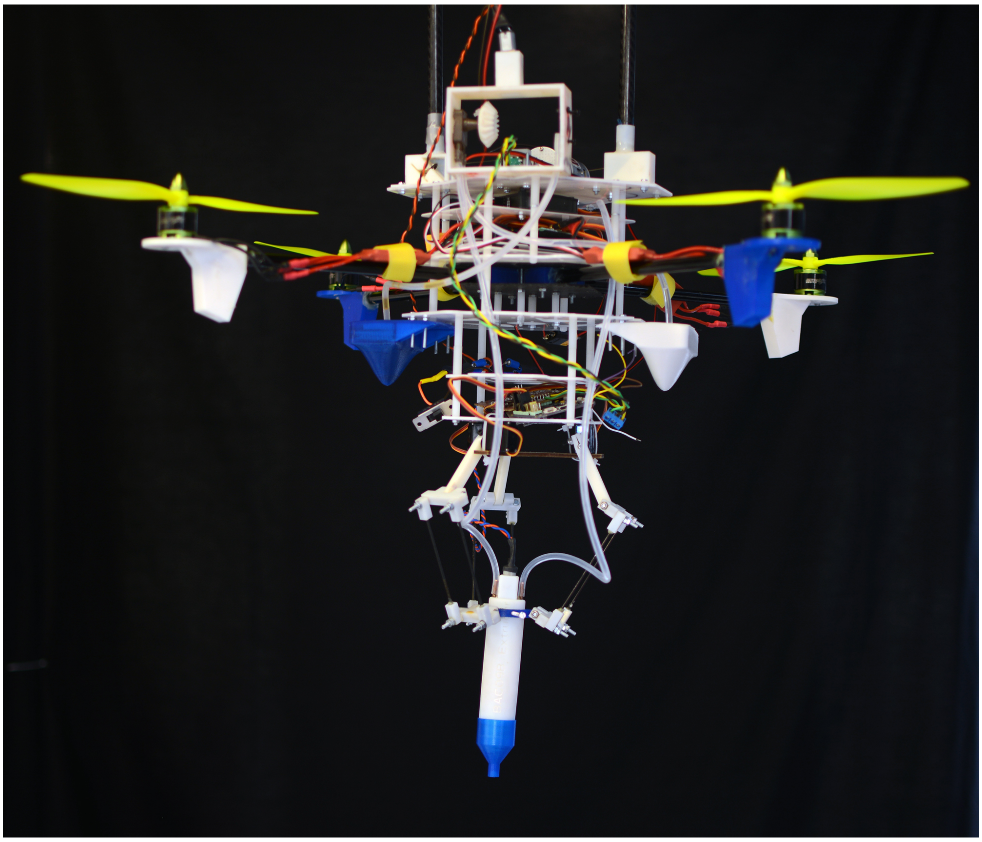


Figure 5.2: Extrusion system mounted on quadrotor

5.2 Full System Testing

Unfortunately due to issues with the flight control system we were unable to test the extrusion system as part of a fully functioning ACV. We were, however, able to build the full system and test the individual subsystems to verify performance. A photo of this is shown in Figure 5.2.

The extrusion control system successfully controlled the perching system, extruder system and motion system as commanded by G-code strings. A serial link was provided simply via USB however in a live system this would be done through the flight controller.

We were expecting to require active stabilisation of the ACV during printing however we were pleasantly surprised with the performance of the perching mechanism. The quadrotor remained very still throughout testing providing an excellent platform for the extrusion system to operate from. The slow and smooth movements of the delta robot coupled with its low inertia design no doubt contributed to this stability.

A short test print was completed on the eve of the submission deadline for this report. All systems functioned correctly and a video of this test can be found accompanying this report.

6 Conclusions

As a simple form of design analysis we can compare our final solution to the objectives set out at the beginning of this report. We stated that our system must:

"process the raw material into an extrudable substance"

The extruder system successfully mixed the two part polyurethane using an active mixing nozzle. The output from this nozzle was a viscous foam which held its shape well and allowed for extrusion on top of previous layers.

"deposit material in a controlled and predictable manner through an end effector"

The foam deposited from the active mixing nozzle was consistent throughout long prints and in various different tests. Furthermore, the output flow rate was slow and controlled enough that the time taken to complete a *lap* was sufficient that the previous layer had hardened enough to support a subsequent layer. One drawback, however, is that it was not possible to stop extrusion once it has started meaning that structures had to be continuous.

"allow the end effector to follow a toolpath and produce complex structures"

The delta robot and motion control system was dexterous and of high enough accuracy that a G-code defined toolpath could be followed to a resolution far exceeding the extrusion width. The low cost servomotors, however, had low quality rotary potentiometers in them for positional feedback. We believe that this caused the servomotors to *jitter* and could be improved using higher quality servomotors or, better yet, a stepper motor with an encoder. Further accuracy could be gained by replacing the 3D printed *ball joints* with a higher quality commercial alternative.

"compensate for ACV motion and keep the end effector on the defined toolpath"

Unfortunately this was not implemented as we were unable to communicate with the OptiTrack motion capture system within the timespan of this project. Further research into this would definitely be advantageous for future ACVs.

"control the perching mechanism to allow for z-axis authority during extrusion"

The MOSFET H-Bridge motor drive circuit functioned excellently in conjunction with the perching mechanism to allow bi-directional high voltage motor drive from a logic level processor. Feedback from the encoder was easily high enough resolution and accuracy to negate the need for any input from a motion capture system. In future it would be more simple and reliable, however, to use a motor drive IC such as the ST MicroElectronics VNH5019.

In general we consider our role in this project to have been successful however it is a shame that, at the time of submission, we do not it have a tangible product of the final solution at full scale.

A Control System Circuit Diagrams

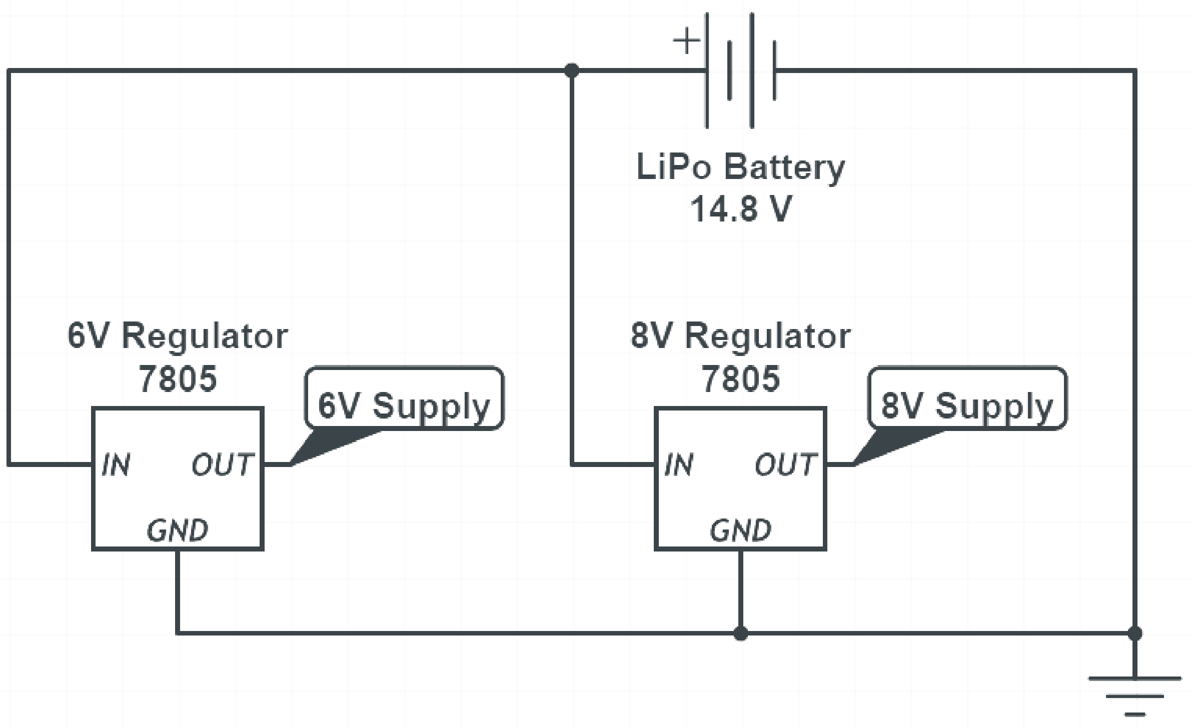


FIGURE A.1: Voltage regulator circuit diagram

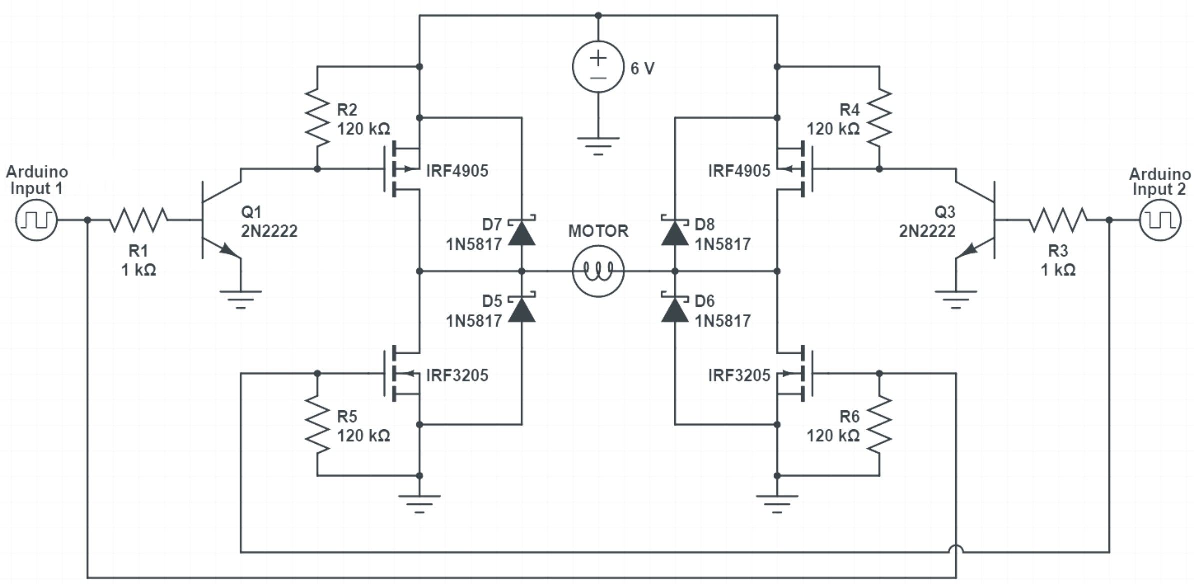


FIGURE A.2: MOSFET H-bridge motor drive circuit

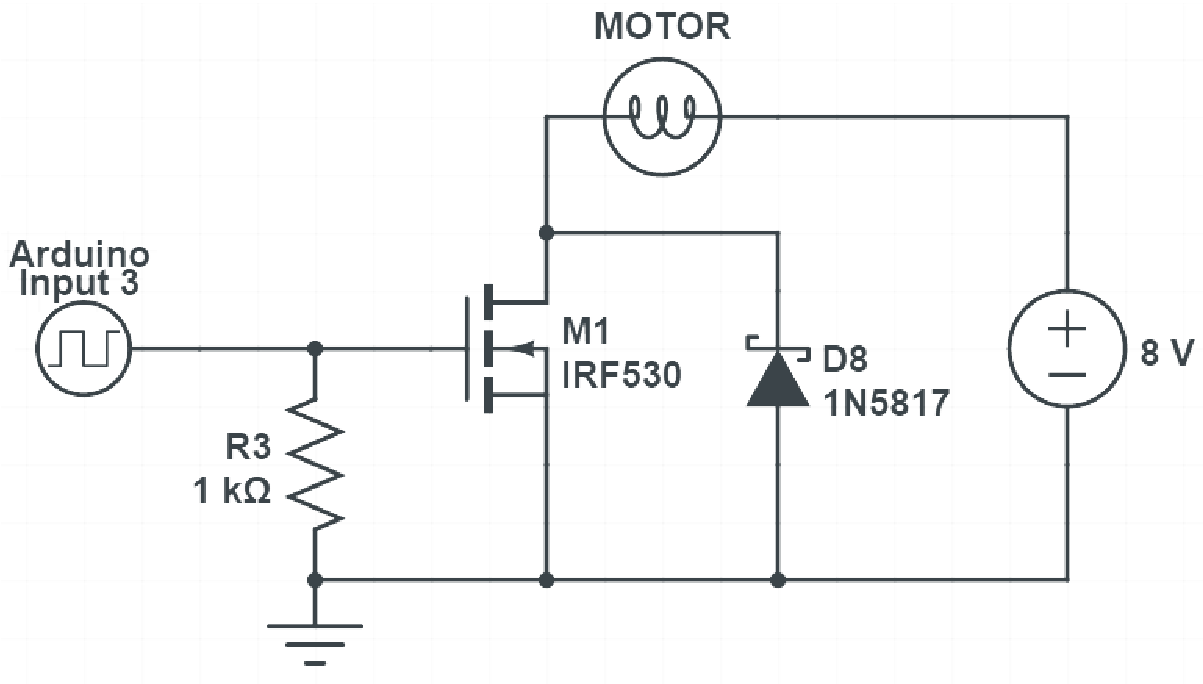


Figure A.3: MOSFET motor driver circuit

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